Practice Set 1 Paper 2 – Urology

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Practice paper compiled from the question bank.

Q1. Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain

Type: LAQ (25 marks) (mHSPC systemic therapy also answered in 2025-Winter Paper 3, Q3b)

Introduction

- This is newly diagnosed (synchronous) **mHSPC (metastatic hormone-sensitive prostate cancer)** with two urgent problems: an obstructed outlet and painful bone metastases.
- Treat the emergencies first, start castration safely (no flare), then intensify systemic therapy.

Immediate assessment

- **Ask and examine for cord compression:** back pain, leg weakness, sensory level, bowel symptoms. DRE (digital rectal examination).
- **Bloods:** PSA (prostate-specific antigen), creatinine, haemoglobin, calcium, ALP (alkaline phosphatase), testosterone.
- **Imaging:** ultrasound for hydronephrosis and residual; **bone scan and CT (computed tomography)**; PSMA (prostate-specific membrane antigen) PET is more sensitive but trials used conventional imaging. **MRI (magnetic resonance imaging) of the whole spine** if any back pain or neurological sign.
- **Histology:** prostate biopsy (or TURP (transurethral resection of the prostate) chips). **Germline genetic testing** for all men with metastatic disease.

Step 1: Relieve the retention

- **Urethral catheter** (suprapubic if urethral fails). Watch for post-obstructive diuresis.
- Persisting renal impairment with hydronephrosis means ureteric obstruction by tumour at the trigone: **nephrostomy or stent**.
- ADT (androgen deprivation therapy) shrinks the prostate, so try **TWOC (trial without catheter)** after a few months of ADT.
- If TWOC fails or symptoms persist: **channel (palliative) TURP**. ADT alone does not prevent local progression (older series: about a third needed TURP within 60 days of orchidectomy).

Step 2: Start ADT at once, without flare

Option	Point
Bilateral orchidectomy	Castrate within hours; best for impending cord compression
GnRH (gonadotropin-releasing hormone) antagonist: degarelix 240 mg subcutaneously, then 80 mg monthly; oral relugolix	**No flare**; castrate by day 3 in >95% (degarelix)
LHRH (luteinising hormone-releasing hormone) agonist + antiandrogen	Agonist raises LH (luteinising hormone) up to 10-fold for 10–20 days (**flare**). Give bicalutamide 50 mg daily with or before the agonist, for 21–28 days

- **Why avoid flare:** the testosterone surge can worsen bone pain, retention and cord compression in spinal metastases.
- Target castrate testosterone **<50 ng/dL**. Antiandrogen alone is not recommended.

Step 3: Bone pain

- **Analgesia** by the WHO (World Health Organization) ladder; neuropathic pain needs gabapentinoids or tricyclics.
- **Focal pain: EBRT (external beam radiotherapy) single 8 Gy fraction.** Pain relief equals multiple fractions; retreatment is commoner.
- X-ray or CT of painful weight-bearing sites to exclude lytic lesions and impending fracture; **orthopaedic fixation** then RT (radiotherapy).
- **Diffuse pain:** steroids, wide-field RT. Radium-223 is licensed only in castration-resistant disease.
- **Bone agents:** denosumab 60 mg 6-monthly or zoledronic acid yearly protect against ADT osteoporosis. SRE (skeletal-related event) doses (zoledronic acid 4 mg or denosumab 120 mg 4-weekly) are for CRPC (castration-resistant prostate cancer). Give calcium, vitamin D and a dental check (osteonecrosis of the jaw).

Spinal cord compression: watch constantly

- Every focal back pain is cord compression until proven otherwise.
- **Dexamethasone** at once (load 10 mg intravenously), urgent whole-spine MRI, then **RT ± decompressive surgery**. Surgery plus RT beats RT alone (**Patchell, 2005**).
- Castrate with orchidectomy or antagonist, never an agonist alone.

Step 4: Intensify systemic therapy (mHSPC)

- ADT alone is no longer standard. Classify volume: **CHAARTED high volume** = ≥4 bone metastases with ≥1 outside the axial skeleton, or visceral metastasis.

Add to ADT	Trial (overall survival HR (hazard ratio))
Abiraterone 1000 mg + prednisone 5 mg	LATITUDE (0.62); STAMPEDE arm G
Enzalutamide or apalutamide (ARPIs (androgen receptor pathway inhibitors))	ENZAMET (0.67); TITAN (0.67)
Docetaxel 75 mg/m² × 6 + abiraterone or darolutamide (**triplet**)	PEACE-1; ARASENS (0.68)

- Hormonal agents work regardless of volume; docetaxel helps only high volume, so it is now given within a triplet.
- **Low-volume synchronous disease:** add **RT to the prostate** (STAMPEDE arm H, 3-year survival 81% vs 73%); it also helps local control.

Follow-up

- PSA, liver tests and blood count every 3–6 months; testosterone if PSA rises. Rising PSA with castrate testosterone = CRPC.
- PSA nadir ≤0.2 ng/mL at 7 months predicts long survival (SWOG 9346).
- Manage ADT effects: bone loss (DEXA (dual-energy X-ray absorptiometry)), diabetes, cardiovascular risk, hot flushes.

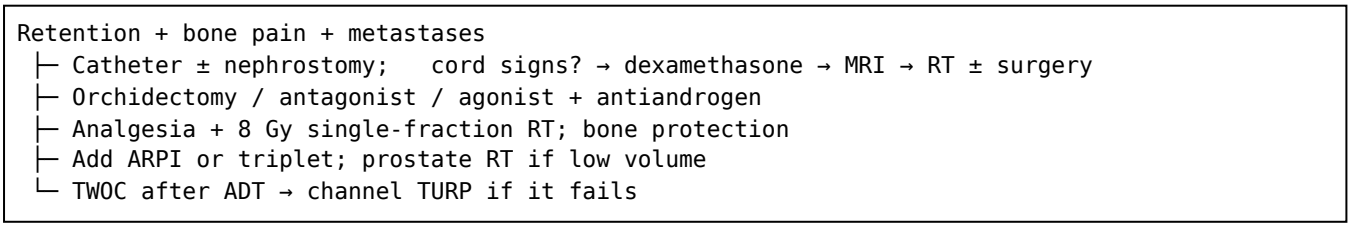


Diagram to draw: the flowchart above.

Key points to remember:

- Never start an LHRH agonist alone with spinal metastases: cover the flare or use an antagonist.
- Single 8 Gy fraction gives the same pain relief as longer courses.
- Focal back pain = cord compression until MRI proves otherwise.
- ADT plus abiraterone or an ARPI, or a triplet, for every fit man.

Q2. Describe neuromuscular dysfunction and management of bladder dysfunction following cervical spine surgery in young adult male

Type: LAQ (25 marks)

Assumed meaning: cervical spinal cord injury (traumatic, operated, or from the surgery itself), giving a suprasacral lesion above T6.

Introduction

- A cervical cord lesion lies **between the PMC (pontine micturition centre) and the sacral cord**. It causes tetraplegia and the classic suprasacral pattern.
- Complete transection is rare; incomplete lesions are commonest. Manage by **urodynamics, not neurology**.

Pathophysiology

- The PMC normally coordinates detrusor contraction with sphincter relaxation. Cutting the descending pathway leaves a **spinal reflex**.
- **Why NDO (neurogenic detrusor overactivity) appears:** unmyelinated C-fibre afferents become sensitised (NGF (nerve growth factor)), sprout and drive a new sacral reflex. Bulbospinal inhibition is lost.

Phase	Detrusor	Outlet	Clinical
Spinal shock (usually 6–12 weeks; up to 1–2 years)	Areflexic, smooth bladder	Bladder neck closed; reduced closure pressure	**Retention**; overflow only
Recovery (suprasacral pattern)	**NDO**, no sensation below lesion	**Striated DSD (detrusor–sphincter dyssynergia)**; lesion above T7–T8 adds **smooth sphincter dyssynergia**	Reflex leakage, incomplete emptying, high pressure

- **Blaivas DSD types:** type 1, sphincter relaxes at peak pressure (incomplete lesions); type 2, sporadic contractions; type 3, crescendo–decrescendo obstruction (types 2–3 with complete lesions).
- **Consequences:** high storage and voiding pressure, VUR (vesicoureteric reflux) in 17–25%, hydronephrosis, stones (about 7% in 10 years), UTI (urinary tract infection). Untreated, about half of men with DSD deteriorate (worse if complete, continuous DSD).
- Upper tract risk also rises with low compliance (<12.5 mL/cm H_2O) and NDO for over a third of filling.
- **VUR:** treat the bladder first (lower the pressure); reflux surgery in a thick bladder is difficult.
- **Also know:** cervical myelopathy may instead give an underactive detrusor.

Autonomic dysreflexia

- Occurs with lesions **above T6** (cervical 60%), after spinal shock. Bladder or bowel distension triggers unopposed spinal sympathetic outflow, as medullary inhibition is lost.
- **Features:** pounding headache, severe hypertension (systolic rise ≥20 mmHg defines it at urodynamics), bradycardia, sweating and flushing above the lesion. It can cause stroke or seizures.
- **Treatment:** sit up, **drain the bladder** (check catheter), then **nitroglycerin 2% paste** above the lesion; intravenous nitroprusside or labetalol if persistent. Prophylaxis with

terazosin or prazosin. Monitor blood pressure during cystoscopy and urodynamics; use spinal or general anaesthesia.

Evaluation

- History, examination, **hand function** (decides self-catheterisation), transfers, carers.
- Urinalysis, creatinine, **renal ultrasound**. Men with good hand function can self-catheterise; others need carers or a suprapubic catheter.
- **VUDS (videourodynamics) after spinal shock**, with blood pressure monitoring: compliance, NDO, DSD, VUR, **DLPP (detrusor leak point pressure) ≥40 cm H_2O** = upper tract risk.
- AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2021: suprasacral lesions are "unknown risk" until tested; DSD or poor compliance = high risk.

Management

Goals: protect kidneys, low-pressure storage, low-pressure emptying, no infection, continence, social acceptability. Start with the simplest reversible option.

- **Spinal shock: CIC (clean intermittent catheterisation)** preferred; indwelling catheter early is acceptable. Defer irreversible procedures.

Aim	Options
Emptying	**CIC: gold standard**, volumes <500 mL; carer CIC in tetraplegics. **Suprapubic catheter** better than long-term urethral catheter. Credé/Valsalva not recommended
Storage (lower pressure)	**Antimuscarinics** ± mirabegron (β3 agonist); **intradetrusor onabotulinumtoxinA 200 U** if refractory (repeat when the effect wears off)
Refractory small bladder	**Augmentation ileocystoplasty** ± continent catheterisable channel (Mitrofanoff)
Men unable to do CIC	**External sphincterotomy** or intrasphincteric botulinum toxin → reflex voiding into condom catheter. High failure; DLPP ≥40 cm H_2O afterwards = failure. Urethral stents abandoned
Neuromodulation	**SARS (sacral anterior root stimulator, Brindley) + posterior sacral rhizotomy**: abolishes NDO and dysreflexia, allows voiding on demand
Last resort	Ileovesicostomy; bladder neck closure + suprapubic catheter; ileal conduit

- **Also know:** sacral neuromodulation is used for neurogenic bladder without SCI (spinal cord injury); hydrophilic catheters delay first UTI.

Complications and follow-up

- **UTI:** treat only symptomatic bacteriuria (5–14 days); no routine prophylaxis.
- Stones, VUR (treat the bladder first), bladder cancer (no screening cystoscopy, AUA/SUFU 2021).
- Lifelong: yearly review and renal function, upper tract imaging every 1–2 years, repeat urodynamics if anything changes.

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Cervical SCI → spinal shock (retention): CIC
   → VUDS at 3 months: NDO + DSD
   → antimuscarinic + CIC → Botox → augmentation
   → cannot CIC: sphincterotomy + condom / SARS / ileovesicostomy
```

Diagram to draw: neural axis: PMC, lesion at cervical cord, sympathetic T10–L2, sacral S2–S4, with the resulting NDO + DSD.

Key points to remember:

- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Q3a. CAPD

Type: SAQ (10 marks) (also answered in 2018-Summer Paper 2, Q3e)

Definition and principle

- CAPD (continuous ambulatory peritoneal dialysis): home dialysis using the **peritoneum** as the membrane.
- **Diffusion** removes urea, creatinine and potassium into the dialysate. **Ultrafiltration** removes water by osmosis: **glucose** (1.5%, 2.5%, 4.25%) or **icodextrin** for the long dwell.
- About **4 exchanges of 2 L daily**, 4–6-hour dwells. APD (automated peritoneal dialysis) uses a night cycler.

Catheter

- **Tenckhoff catheter:** silicone, **two Dacron cuffs** (deep cuff in rectus; superficial cuff near the exit), downward exit, tip in the pelvis.
- Insertion: open, **laparoscopic** (rectus sheath tunnel, omentopexy, adhesiolysis) or percutaneous. Delay full volumes about 2 weeks to prevent leaks.

Adequacy

- Weekly **Kt/V urea ≥1.7** (urea clearance × time ÷ urea distribution volume) plus clinical well-being. PET (peritoneal equilibration test) identifies high transporters, who suit short dwells.

Advantages over haemodialysis

- Home-based, no vascular access, **haemodynamically gentle**, **preserves residual renal function**. Favoured in children, unstable heart disease and poor vascular access.

Contraindications

- **Absolute:** peritoneal scarring or adhesions, membrane failure, uncorrectable hernia, nobody to do exchanges.
- **Relative:** recent abdominal surgery, stoma, polycystic kidneys, morbid obesity.

Complications

- **Peritonitis** (2 of 3): pain or **cloudy effluent**; effluent **WCC (white cell count) >100/µL, >50% neutrophils**; positive culture. Commonest: coagulase-negative *Staphylococcus*. Give **intraperitoneal** Gram-positive and Gram-negative cover; remove catheter if fungal or refractory (5 days).
- **Exit-site and tunnel infection** (topical mupirocin).
- **Mechanical:** outflow failure (constipation, omental wrap), leak, **hernia**, hydrothorax.
- **Metabolic:** hyperglycaemia, weight gain, protein loss.
- **Long term:** ultrafiltration failure; **EPS (encapsulating peritoneal sclerosis)**.
- **Also know:** remove the PD catheter after a successful transplant; retroperitoneal laparoscopy protects the peritoneum for PD (peritoneal dialysis) patients.

Diagram to draw: Tenckhoff catheter with two cuffs, tunnel, downward exit, pelvic tip.

> **Key points to remember:**
> - Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
> - Two-cuff Tenckhoff catheter; Kt/V ≥1.7 weekly.
> - Peritonitis: cloudy fluid, WCC >100/µL with >50% neutrophils.

Q3b. Flexible URS

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 2, Q3b; digital scopes in 2019-Summer Paper 4, Q3c)

Introduction

- Flexible URS (ureteroscopy), or RIRS (retrograde intrarenal surgery), treats the whole upper tract by the natural route. Marshall first reported it; a working channel and active deflection made it therapeutic.

Instrument

Feature	Details
Deflection	**≥270° both ways** (reaches lower pole)
Working channel	**3.6 Fr** (laser fibre, basket)
Fibreoptic	Smallest tip; fibres break → black spots; repair about every 15 cases
Digital ("chip-on-tip", CMOS (complementary metal-oxide semiconductor))	Better image; larger tip (about 8.5 Fr); not more durable
Single-use (LithoVue, 2016)	No repair or reprocessing; cheaper below about 99 cases/year

Indications

- **Stones:** renal <2 cm; lower pole <1 cm (or 1–2 cm as PCNL (percutaneous nephrolithotomy) alternative); **must continue anticoagulants**; failed SWL (shock wave lithotripsy); obesity.
- **Others:** diagnosis and laser ablation of low-grade UTUC (upper tract urothelial carcinoma); strictures, endopyelotomy, calyceal diverticulum, haematuria.

Technique

1. Sterile urine; single-dose antibiotic; α-blocker or **pre-stenting** if the ureter is tight.
2. Cystoscopy, **safety guidewire**, retrograde pyelogram.
3. **UAS (ureteral access sheath)**, 12/14 Fr: lowers intrarenal pressure and temperature, eases repeated passes, but injures the ureter in nearly half (severe in about 1 in 8).
4. Low-pressure irrigation. **Holmium:YAG (yttrium-aluminium-garnet) or thulium fibre laser:** fragment and basket, or dust. Keep power ≤10 W in ureter, ≤25 W in kidney (heat above 43 °C damages tissue).
5. Relocate lower pole stones to an upper calyx.
6. Inspect on withdrawal. **Omit stent** if normal anatomy, normal renal function, no injury and no second stage.

Complications

- Stent symptoms (commonest), **sepsis about 5%**, stricture about 3%, perforation, bleeding <1%, **avulsion <1%** (nephrostomy, delayed repair), scope damage.

Recent advances

- Pressure-sensing scopes, suction sheaths, thulium fibre laser, robotic URS, AI (artificial intelligence) navigation.

Diagram to draw: scope through access sheath, deflected into lower pole calyx.

> **Key points to remember:**
>
> - 270° deflection, 3.6 Fr channel.
> - Access sheath: lower pressure, but ureteric injury.
> - Modality of choice when anticoagulants cannot stop.

Q3c. Anomalies of renal artery.

Type: SAQ (10 marks)

Normal pattern

- One renal artery from the aorta at L1–L2; anterior and posterior divisions; **five segmental end arteries** (apical, upper, middle, lower, posterior). The right artery passes behind the IVC (inferior vena cava).

Terminology

Term	Definition
Multiple renal arteries (preferred)	More than one vessel from the aorta to one kidney
Aberrant	Arises from a vessel other than aorta or main renal artery (iliac, SMA (superior mesenteric artery), gonadal)
Accessory	Two or more branches supplying the same segment

Incidence

- Single artery in **71–85%**; variants in 25–40%. Accessory arteries 25–28% (aorta T11–L4), usually to the **lower pole**. True aberrant vessels are rare except in **ectopic and horseshoe kidneys**.
- **Also know:** early prehilar branching; right artery anterior to IVC in about 5%.

Embryology

- The ascending kidney takes successive aortic branches; the lower ones normally degenerate. **Failure of degeneration** gives polar or multiple arteries.

Clinical significance

- **PUJ (pelviureteric junction) obstruction** from a lower-pole crossing vessel; but crossing vessels occur in a fifth to a quarter of normal PUJs.
- Compression of an infundibulum: hydrocalyx, pain, stones, UTI (urinary tract infection).
- **Surgery:** bleeding at endopyelotomy (incise posterolaterally); polar arteries are end arteries, so ligation infarcts that segment; **donor nephrectomy** needs CT (computed tomography) angiography; aortic surgery.
- Not linked to hypertension.

Renal artery aneurysm

- About **0.1%**. **Abeshouse types:** saccular (about 93%), fusiform, dissecting, arteriovenous.
- Mostly asymptomatic; hypertension, haematuria, pain; wreath-like calcification.
- **Treat** (endovascular or open) if rupture, **>2 cm**, expanding, **pregnancy or planned pregnancy**, symptoms or emboli.

Renal arteriovenous fistula

	Congenital (cirsoid AVM (arteriovenous malformation))	Acquired
Share	<25%	Majority (biopsy, PCNL (percutaneous nephrolithotomy), trauma)
Features	Bruit, haematuria, high-output failure	Often resolves
Treatment	Selective embolisation	Observe; embolise if bleeding

Diagram to draw: kidney with main artery, lower-pole accessory artery crossing the PUJ.

> **Key points to remember:**
>
> - Single artery in 71–85%; accessory arteries usually lower pole.
> - Crossing vessel can cause PUJ obstruction.
> - Treat aneurysm >2 cm or in pregnancy.

Q3d. Pheochromocytoma

Type: SAQ (10 marks) (pre-operative management in detail: 2020-Summer Paper 2, Q3c)

Definition and epidemiology

- Catecholamine-secreting tumour of adrenal medullary chromaffin cells; **paraganglioma** is the extra-adrenal form (organ of Zuckerkandl, bladder).
- 1–2 per 100,000 a year; about 5% of adrenal incidentalomas.
- **"Rule of 10" is outdated:** extra-adrenal up to 25%, **hereditary about one-third**, bilateral commoner in hereditary disease.

Genetics

- **MEN2 (multiple endocrine neoplasia type 2)**, RET: about 50% risk, adrenal, low malignancy. **VHL (von Hippel–Lindau)**: normetanephrine only. **NF1 (neurofibromatosis type 1)**. **SDHB (succinate dehydrogenase subunit B):** extra-adrenal, **30–50% metastatic**.
- Offer genetic testing to all.

Clinical features

- Triad: **headache, sweating, palpitations**. Sustained hypertension is commoner than paroxysmal; pallor, hyperglycaemia; >20% asymptomatic. Epinephrine-dominant tumours can cause hypotensive episodes.

Diagnosis

- **Biochemistry first:** plasma free or 24-hour urinary fractionated **metanephrines**. Tumours convert catecholamines continuously, so metanephrines are more sensitive than episodic catecholamines. Chromogranin A; clonidine suppression for borderline results.
- **Imaging:** CT (computed tomography): >10 HU (Hounsfield units) unenhanced, no rapid washout; MRI (magnetic resonance imaging). **Functional:** ^{68}Ga-DOTATATE PET (positron emission tomography), FDG (fluorodeoxyglucose) PET (best for SDHB), ^{123}I-MIBG (metaiodobenzylguanidine) (best in MEN2).
- **Never biopsy** (hypertensive crisis). Iodinated contrast is safe.

Pre-operative preparation (7–14 days)

- **α-blockade for all:** phenoxybenzamine 10 mg twice daily, titrated, or doxazosin; calcium channel blocker as alternative.
- **β-blocker only after α-blockade**, because unopposed α-stimulation causes crisis.
- **High salt and fluids**; target blood pressure 120–130/80 mmHg with mild postural drop. Echocardiography.

Surgery

- **Laparoscopic adrenalectomy** (open for large or invasive); minimal handling; early adrenal vein ligation. **Cortical-sparing** partial adrenalectomy in MEN2/VHL.
- Post-op: **hypotension** and **rebound hypoglycaemia**; HDU (high dependency unit) monitoring.

Metastatic disease and follow-up

- Malignancy is defined only by **metastases**. Treat with metastasectomy, ^{131}I-MIBG, CVD (cyclophosphamide, vincristine, dacarbazine).
- **Lifelong yearly metanephrines** (recurrence up to 16%).

Diagram to draw: catecholamine pathway to metanephrines (COMT (catechol-O-methyltransferase)).

Key points to remember:

- Metanephrines first; never biopsy.
- α before β, plus volume expansion.
- One-third hereditary; SDHB most often metastatic.

Q3e. Microscopic Hematuria

Type: SAQ (10 marks) (also answered in 2016-Summer Paper 3, Q3d)

Definition

- **MH (microscopic haematuria): ≥3 RBCs/HPF (red blood cells per high-power field)** on microscopy of one properly collected sample (AUA/SUFU (American Urological Association/Society of Urodynamics, Female Pelvic Medicine and Urogenital Reconstruction) 2020).
- **Dipstick alone is insufficient**: myoglobin, semen and bacterial peroxidases give false positives.
- Prevalence about 6.5%; malignancy in about 3% (over 85% bladder cancer).

Causes

- Malignant: bladder, UTUC (upper tract urothelial carcinoma), RCC (renal cell carcinoma), prostate.
- Benign: UTI (urinary tract infection), stones, BPE (benign prostatic enlargement), stricture, TB (tuberculosis), schistosomiasis.
- **Medical renal:** IgA nephropathy (dysmorphic RBCs, casts, proteinuria).
- Anticoagulants do not explain haematuria: evaluate as usual.

Initial evaluation

- History (smoking, occupation, cyclophosphamide, pelvic radiation, Lynch syndrome), examination, **blood pressure, creatinine**.
- Suspected UTI → culture, treat, **repeat urinalysis**; persistent MH → full evaluation.
- Renal features → nephrology **and** urological evaluation.

AUA/SUFU 2020 risk stratification

	Low (all)	Intermediate (any)	High (any)
Age	Women <50; men <40	Women 50–59; men 40–59	≥60
Smoking (pack-years)	Never or <10	10–30	>30
RBCs/HPF	3–10	11–25	>25
Gross haematuria history	No	No	**Yes**
Cancer risk	0–0.4%	1–2.5%	About 6%
Work-up	Shared decision: repeat urinalysis in 6 months **or** cystoscopy + RUS (renal ultrasound)	Cystoscopy + RUS	Cystoscopy + **CT (computed tomography) urography**

- Additional risk factors (irritative symptoms, radiation, cyclophosphamide, occupational exposure) move low to intermediate.
- CT contraindicated → MR (magnetic resonance) urography, or retrograde pyelograms.
- **No cytology or urine markers initially**; white-light cystoscopy.
- Persistent MH after a negative ultrasound → CT or retrograde imaging. Hereditary RCC risk (VHL (von Hippel–Lindau)) → upper tract imaging in any stratum.

Follow-up

- Negative evaluation → consider repeat urinalysis within 12 months; release if negative.
- Re-evaluate for gross haematuria, rising MH or new symptoms.

Diagram to draw: AUA/SUFU flowchart from MH to risk-based work-up.

> **Key points to remember:**
> - ≥3 RBCs/HPF on microscopy; dipstick not enough.
> - Intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
> - Any gross haematuria = high risk.

Q3f. PIN

Type: SAQ (10 marks)

Assumed meaning: prostatic intraepithelial neoplasia.

Definition

- PIN (prostatic intraepithelial neoplasia): **architecturally benign** acini or ducts lined by **cytologically atypical** cells, with an intact or patchy **basal cell layer**.
- Graded **LGPIN (low-grade PIN)** and **HGPIN (high-grade PIN)**. LGPIN is not reported: it is irreproducible and carries no excess risk.

Histology

- HGPIN: nuclear enlargement, hyperchromasia, **prominent nucleoli**.
- Patterns: tufting, micropapillary, flat. **Cribriform lesions are now classed as IDC-P (intraductal carcinoma of the prostate) or AIP (atypical intraductal proliferation).**
- Immunohistochemistry: basal cells stain with **HMWCK (high-molecular-weight cytokeratin) and p63**; AMACR (α-methylacyl-CoA racemase) positive. Mainly in the peripheral zone.

Evidence that HGPIN is a precursor

- More and larger HGPIN foci in cancerous prostates; carcinoma seen budding from HGPIN.
- About 20% carry **TMPRSS2:ERG fusion** (about 50% of cancers). Extra genomic changes are needed for invasion.

Incidence

- About **4–5%** of biopsies (range 0–25%, due to observer variation).

Related lesions

Lesion	Meaning	Action on isolated finding
HGPIN	Weak risk marker	**No repeat biopsy**
PINATYP (HGPIN with adjacent atypical glands)	As "atypical"	Manage as ASAP (atypical small acinar proliferation)
ASAP	Cancer on re-biopsy about 40%	Repeat biopsy ± MRI (magnetic resonance imaging)
AIP	May herald IDC-P	Consider re-biopsy
IDC-P	Usually high-grade cancer present	Definitive treatment; excludes active surveillance

Management

- Isolated HGPIN, unifocal or multifocal: **no immediate repeat biopsy**. Decide further testing on PSA (prostate-specific antigen), DRE (digital rectal examination) and **mpMRI (multiparametric MRI)**.
- HGPIN on TURP (transurethral resection of the prostate) chips: no further sampling on current data.
- **Chemoprevention:** none proven; green tea showed reduced progression in small trials.
- *Guideline note:* EAU (European Association of Urology) guidelines still list extensive HGPIN (≥3 biopsy sites) as a reason for repeat biopsy; the chapter does not.

Diagram to draw: benign gland, HGPIN gland (atypical cells, intact basal layer) and cancer (no basal layer).

Key points to remember:

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.

Rapid Revision Sheet – Practice Set 1 Paper 2

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Treatment of metastatic CA prostate in 60 yr male with retention of urine and bone pain (LAQ)

- Never start an LHRH agonist alone with spinal metastases: cover the flare or use an antagonist.
- Single 8 Gy fraction gives the same pain relief as longer courses.
- Focal back pain = cord compression until MRI proves otherwise.
- ADT plus abiraterone or an ARPI, or a triplet, for every fit man.

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- Suprasacral lesion = NDO + striated DSD; above T7–T8 also smooth sphincter dyssynergia.
- Autonomic dysreflexia above T6: drain bladder first, nitrates next.
- CIC + bladder relaxation (antimuscarinic, Botox 200 U) is the mainstay.
- DLPP ≥40 cm H_2O threatens the kidneys.

Q3a. CAPD (SAQ)

- Peritoneum is the membrane; glucose or icodextrin drives ultrafiltration.
- Two-cuff Tenckhoff catheter; Kt/V ≥1.7 weekly.
- Peritonitis: cloudy fluid, WCC >100/µL with >50% neutrophils.

Q3b. Flexible URS (SAQ)

- 270° deflection, 3.6 Fr channel.
- Access sheath: lower pressure, but ureteric injury.
- Modality of choice when anticoagulants cannot stop.

Q3c. Anomalies of renal artery. (SAQ)

- Single artery in 71–85%; accessory arteries usually lower pole.
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- Treat aneurysm >2 cm or in pregnancy.

Q3d. Pheochromocytoma (SAQ)

- Metanephrines first; never biopsy.
- α before β, plus volume expansion.
- One-third hereditary; SDHB most often metastatic.

Q3e. Microscopic Hematuria (SAQ)

- ≥3 RBCs/HPF on microscopy; dipstick not enough.
- Intermediate: cystoscopy + ultrasound; high: cystoscopy + CT urography.
- Any gross haematuria = high risk.

Q3f. PIN (SAQ)

- Atypical cells, benign architecture, basal layer retained.
- Isolated HGPIN needs no repeat biopsy; ASAP does.
- Cribriform "PIN" is now IDC-P or AIP.